



Karsten Held

Azure AI Engineer
Azure, M365, Enterprise RAG,
GenAI Solutions

Karsten Held
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Personal Profile

AI Engineer with 25+ years in software development and 3+ years shipping enterprise GenAI solutions at scale. Designed and shipped a market intelligence RAG platform achieving 90% factuality (vs 53% for Microsoft Copilot) at 1/3 the cost. Built enterprise search crawler pipelines rolled out to 5,000+ users, ingesting SharePoint content across multiple sites and domains. Full-stack sole-developer delivery: Python backends (Azure Functions, FastAPI), TypeScript/React frontends (SPFx), Azure AI infrastructure. Security hardening (Managed Identities, RBAC), automated evaluation, and production deployment. Fluent in German, English, and Hungarian.

Available for freelance/contractor roles in AI Engineering, and Azure / SharePoint development starting August 2026.

Areas of Expertise

- **Enterprise RAG & GenAI:** Responses API, Vector Stores, metadata extraction, automated evaluation
- **Azure AI Infrastructure:** Azure OpenAI Service, Foundry IQ (Azure AI Search), Azure Functions, Managed Identities, RBAC
- **Crawler & Data Pipelines:** SharePoint crawling, content embedding, overnight sync, multi-source ingestion
- **Frontend & Integration:** TypeScript, React, Fluent UI, SPFx 1.23, streaming API, multi-tenant deployment

Career History



2024 - 2026-03

External Azure AI Engineer at [Vattenfall](#)

- **2025 - 2026 Enterprise Search SharePoint Connector:** Crawler pipeline ingesting documents, lists, and site pages from 3 SharePoint domains into Azure AI Search. Metadata enrichment via Azure OpenAI (embeddings, markdown conversion). Domain Tag system for scoped retrieval. Managed Identity authentication (Sites.Selected)
- **2024 - 2025 SharePoint GPT Extensions (SPFx 1.20, React, TypeScript):** AI-powered frontend with RAG Chat, File Uploader with AI metadata extraction, List Command Set. 20+ Fluent UI components, streaming API, per-site configuration, multi-tenant deployment, in-app self-test suite.
- **2023 - 2024 Lookout GPT - Market Intelligence Platform:** Custom RAG for Vattenfall strategy team. Q&A over 2000+ competitor reports. OpenAI Responses API + Vector Stores, Azure Functions middleware, overnight crawler, megaprompt optimization (4x token reduction). Sole developer, multi-site deployment.

2016 - 2024

External SharePoint / Power Platform Specialist at [GASAG](#), [Vattenfall](#), [DHL](#), [Telekom](#)

- **2022 - 2024 SharePoint Site Provisioning App:** SPFx frontend + Dataverse backend, 10k+ managed sites, row-level security at scale, Power Automate lifecycle automation, template-based provisioning with approval workflows
- **2023 - 2024 Change Management Request PowerApp:** Canvas app with multi-level approval workflows, 7-step form wizard, custom PCF components (TypeScript/React/Fluent UI), Power Automate integration
- **2019 - 2024 SharePoint and Power Platform 3rd Level Support:** 5+ model-driven apps, Dataverse backends
- **2016 - 2019 Contact management solution (SharePoint, GASAG),** backend/frontend optimization (C#, Entity Framework, SQL), IT Security Self Assessment (SharePoint, DHL), Site Provisioning (C#, Telekom)

2013 - 2015

SharePoint Consultant at [Communardo](#), [M-Events](#), [AUC](#), [GEOMAGIC](#)

- Intranet solutions (Axel Springer AG, Voestalpine Group), Office 365 migrations, web applications (Vaadin)

2001 - 2008

Application Developer, Junior Fund Manager, Team Lead at [Vescore Solutions AG](#)

- 2006 - 2008 Developed central Asset Management Infrastructure (C#, VBA, Excel, Bloomberg)
- 2001 - 2006 App Development (Java, C#), 3 years Junior Fund Manager (Value, Private Equity)

2000

Software Developer at [Dr. Grob & Partner AG](#), Winterthur

- Developed Bridge and Tunnel Management System (Visual Basic 6, ORACLE)

Education

Microsoft Credentials
2013 - 2025

- Exam AI-102: Designing and Implementing a Microsoft Azure AI Solution
- Exam AI-900: Azure AI Fundamentals
- Exam 331: Core Technologies for Microsoft SharePoint Server 2013
- Exam 486: Developing ASP.NET MVC Web Applications
- Exam 480: Programming in HTML5 with Javascript and CSS3
- Exam 488: Developing Microsoft SharePoint Server 2013 Core Solutions



1994-10 - 2000-02
LANGUAGES

Master of Civil Engineering (Dipl.-Ing.) at [Bauhaus-University Weimar](#)
German, English, Hungarian



Karsten Held

Technology Profile

Azure App Service / Function Apps
Foundry IQ (Azure AI Search)
Azure OpenAI Service

Python / FastAPI
TypeScript
React / Fluent UI

Power Automate
SharePoint / SPFx
C# / .NET / .NET Core

Key Project Highlight

Lookout GPT - Market Intelligence Platform (2024-2025)

Full-stack AI solution for Vattenfall's strategy team. RAG-based conversational Q&A over 2000+ competitor reports using OpenAI Responses API with Vector Stores. Achieved 90% factuality in automated benchmarks vs. 53% for Microsoft Copilot. AI metadata extraction using optimized megaprompts (4x token reduction). Multi-site deployment with overnight crawler, per-site configuration, and security-trimmed access. Cost: 10 EUR/user/month (1/3 of Copilot). Sole developer and architect.

Technologies

Primary (daily use – directly relevant)

- Python 3.12, FastAPI, Azure Functions
- Azure OpenAI Service, Responses API, Vector Stores
- Foundry IQ / Azure AI Search
- RAG: chunking, retrieval, evaluation
- TypeScript, React, Fluent UI, SPFx 1.23
- Managed Identities, RBAC, MSAL.js

Secondary (strong background)

- SharePoint REST API, PnP/SP, Microsoft Graph
- Azure App Service, Key Vault, Microsoft Entra ID
- Power Apps, Dataverse, Power Automate
- C# / .NET, SQL, Entity Framework
- HTML5, CSS3, SCSS Modules, Webpack

Tertiary (available)

- Node.js, PowerShell, Azure CLI
- Azure DevOps Pipelines, Git/GitHub
- httpx, BeautifulSoup4, Playwright

Key Technical Achievements

- 90% factuality** in RAG benchmarks vs 53% for Microsoft Copilot (30 expert test cases, Microsoft Foundry evaluation)
- 4x token reduction** in AI metadata extraction via megaprompt optimization (1.5M → 357K tokens)
- Enterprise crawler pipeline** ingesting documents, lists, and site pages from 3 SharePoint domains
- Full-stack AI delivery:** Python backend + TypeScript/React frontend + Azure infrastructure, sole developer
- Multi-site crawler RAG deployment** rolled out to 5,000+ users with per-site configuration and security-trimmed access
- 20+ Fluent UI / React components** in production AI chat interface with streaming responses

Services



Microsoft Foundry (Azure AI Foundry)

Automated evaluation pipelines, LLM-as-judge scoring, factuality / relevance / coherence metrics, prompt optimizations, evaluations, model comparison.



Microsoft Copilot Ecosystem

Copilot extensibility, Copilot connectors, SharePoint Premium, custom RAG alternatives where Copilot falls short.



Foundry IQ (Azure AI Search)

Custom indexes, hybrid retrieval (keyword + vector), metadata enrichment, SharePoint connectors, Copilot Retrieval API integration.



SharePoint Solutions

Information Architecture, Solution Design, Custom Collaboration Sites, Document Management Systems, Portals, Web Part Development, Integration with Azure Logic Apps or Power Automate.